

OIL INDIA LIMITED · IIT KHARAGPUR

Platinum Jubilee Innovation Hackathon

PRANA

Industrial Energy Flexibility Intelligence

Process Response & Adaptive Network Agent

*A Process-Engineering Digital Twin and AI Optimization
Platform for Safe, Economic Industrial Load Flexibility*

Competition

Oil India Limited / IIT Kharagpur Innovation Challenge

Track

Track 4 Digital Asset Management (AI/ML, Digital Twins, SCADA, Robotics, Corrosion)

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This document distinguishes real settled market data, modelled reference-case results, engineering assumptions, and future deployment requirements throughout. The headline economic figures are annualised modelled economic potential for a reference archetype, not measured savings from an operating plant.

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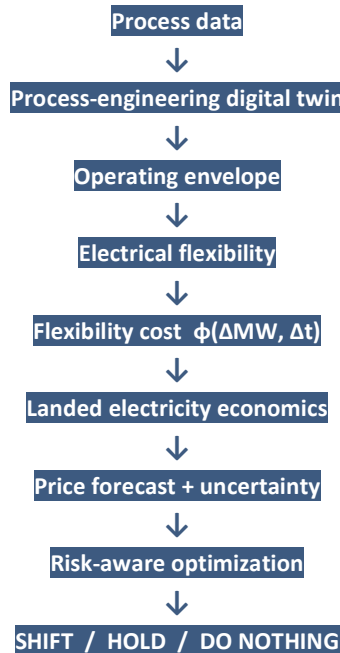
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Executive Summary

Industrial electricity prices in India no longer behave like a flat-cost input. On the real, settled 15-minute Indian Energy Exchange (IEX) panel for the Maharashtra bid area used throughout this proposal 152,542 blocks, 1,588 complete days, April 2022 to August 2026 the ratio of mean evening price to mean solar-hour price rose from 1.47 in FY2022-23 to 2.23 in FY2025-26, and to 3.53 in the Apr–Aug 2026 window of FY2026-27 (seasonally biased upward; labelled wherever it appears). The share of the year clearing below ₹2/kWh grew from 4.1% to 21.1% over the same period.

Industrial processes cannot simply chase this price signal, because every process answers to constraints the market cannot see: safety interlocks, production commitments, equipment ramp limits, co-product balances, inventory bounds, downstream offtake, and utility limits. A price signal alone is not an operating instruction the same ₹/kWh can imply completely different actions depending on what a plant is making and which constraint currently binds.

PRANA (Process Response & Adaptive Network Agent) is a decision-intelligence layer inserted between the electricity market and the plant's own operating envelope. It does not treat the plant as a battery. It *derives* from process physics how much electrical load the plant can safely and economically move, prices that movement in rupees, and only then compares it against what the market is paying. The architecture is a pipeline:



The platform is demonstrated end to end on one archetype: a 700 tonne-per-day (TPD), 67 MW nominal chlor-alkali cell house.

[MODELLED REFERENCE CASE] The 700 TPD plant does not exist as a specific, named site. It is a modelled reference archetype whose parameters sit inside published engineering ranges (National Productivity Council, Government of India, Chlor-Alkali Sector Manual) but have not been fitted to any single plant's historian. Scale mostly cancels in the ₹/kWh headline; it does not cancel in absolute rupee figures.

Chlor-alkali was selected because power is 50–60% of its production cost (NPC, p.22), it buys from the grid rather than running captive generation, and it owns a caustic-soda storage tank that behaves as a state-of-

charge buffer. Replayed over 120 consecutive real delivery days, at real settled prices and with real forecast error, the reference case shows a **modelled** net saving of **₹0.133 per kWh** of total plant load **₹9.17 crore per year annualised** (₹301.5 lakh over 120 days, net of ₹76.8 lakh of membrane wear, diverted chlorine and power-factor cost against a ₹378.3 lakh gross benefit) with zero constraint violations under independent verification against the true nonlinear physics, and 13 of 120 days reported as worse than doing nothing.

[MODELLED REFERENCE CASE] ₹9.17 crore/year is an annualised modelled economic potential for the reference archetype. It is NOT measured savings from an operating plant. No plant has saved this money; the figure is the output of a simulation run on real prices and a real published tariff order against a plant that was designed on paper for this exercise.

The engineering contribution is not the observation that industrial demand response is possible that is an active, published research area (Otashu & Baldea; Weigert et al.; Richstein & Hosseinioun) and is treated as such throughout this document, with PRANA's own numbers benchmarked against it in §22 and §28. The contribution is the *integration*: a process-engineering flexibility cost curve derived from first-principles electrochemistry, dispatched under uncertainty against the full landed cost of Indian industrial power not the exchange price, which is roughly 54% of what an EHV consumer actually pays with a language-model agent used narrowly to extract constraints a plant has never written down, and every recommendation independently re-verified against the true nonlinear physics before it is shown to anyone.

This document sets out the engineering basis for each of these claims, states plainly which numbers are real, which are modelled, which are assumptions with a stated direction of error, and what would be required to calibrate the platform to an operating plant.

1. The Industrial Problem

Two structural changes are colliding in Indian industrial electricity procurement.

1.1 The market has become time-shaped

Grid-scale solar has changed the *shape* of the electricity price within a delivery day, not just its level. On the settled IEX panel behind this proposal, midday prices increasingly approach zero while evening prices repeatedly reach the ₹10/kWh regulatory cap. The evening-to-midday price ratio rose from 1.47× to 3.53× over roughly four years (Apr–Aug 2026 figure is seasonally biased upward and is labelled as such everywhere it is used). A flat industrial load is, by construction, buying an increasing share of its energy at the most expensive hour of the day.

1.2 The bill is not the market price

An industrial consumer at EHV in Maharashtra does not pay the market clearing price. It pays a landed cost built from time-of-day multipliers, wheeling and cross-subsidy surcharges, electricity duty, and usually the largest single non-energy item a demand charge on the single highest billing-demand block of the month. On the verified MERC tariff order used in this proposal (§8), the demand charge alone is **₹22,109 per MW of peak demand, per day**. Roughly 46% of what an industrial consumer pays is not the energy price at all.

1.3 Why price arbitrage alone is insufficient

A tool that reacts only to the exchange price can give actively harmful advice. Shifting load into a cheap midday hour can create a new, higher monthly billing peak and that new peak can cost more than the energy saved. §8 of this document quantifies this trade-off directly for the reference plant.

1.4 The constraints the market cannot see

Every process answers to a set of engineering constraints before it answers to price:

- Safety interlocks hard bounds an optimizer must never cross.
- Production commitments and downstream demand a co-product plant cannot simply idle a unit whose output another unit is consuming in real time.
- Equipment ramp limits and cycling wear moving a setpoint quickly, or often, has a real cost even when the destination setpoint is otherwise free.
- Inventory bounds a storage buffer has finite usable capacity and a required minimum cover.
- Utility and co-product balances many processes produce more than one stream simultaneously, and the second stream may be harder to store than the first.

These constraints are process-engineering facts, not market facts. The market knows the price; the plant knows the process. PRANA's premise is that the missing layer is the connection between the two and that the connection has to be built from real process physics, or its advice is not credible to the people who would have to act on it.

This document does not claim that industrial demand response is a newly invented concept. Time-of-day operation, demand response programmes, energy management systems, and advanced process control are all established practice, discussed on their own terms in §2.

2. Existing Approaches and the Specific Gap

Several categories of tooling already address parts of this problem, and none of them is claimed here to be generally inferior:

- Time-of-day (ToD) operation a manual or rule-based practice, already used informally by many Indian chlor-alkali plants to avoid the worst peak hours.
- Demand-response (DR) programmes utility- or aggregator-run schemes that pay for a pledged MW reduction.
- Energy management systems (Schneider, Siemens, Honeywell and others) these report and visualise consumption; they generally do not model process constraints or trade against a forecast.
- Advanced Process Control / Model Predictive Control (APC/MPC) these optimize the process itself, typically treating power price as a constant or an afterthought.
- Battery energy storage systems (BESS) genuine dispatchable storage, but built at real capital cost.
- Electricity-market optimization tools these typically optimize against the exchange clearing price alone, which (\$1.2) is not what an industrial consumer actually pays.

The specific gap PRANA addresses is narrower than any of the above, and is stated precisely: *determining the plant-specific, state-dependent, economically usable flexibility, after process constraints and process costs are considered, and dispatching it against the landed not exchange cost of power.*

None of the following, individually, is the innovation: a ramp-rate limit is not the innovation; an inventory buffer is not the innovation; demand response itself is not the innovation. The contribution is the integration of:

- process engineering (a derived, not fitted, power-vs-production relationship),
- flexibility quantification (how much load can move, for how long, without breaching a bound),
- flexibility cost (what that movement costs the process itself wear, diverted co-product, efficiency penalty),
- landed electricity economics (what power actually costs at the meter, not at the exchange),
- forecast uncertainty (a price range, not a point forecast),
- risk-aware optimization (a schedule that performs acceptably across that range, not only at its midpoint), and
- operator-readable, solver-derived decision logic (an explanation a plant engineer can check, not a model's unverified narrative).

3. The PRANA Solution

PRANA sits between the electricity market and the plant's existing control systems as an advisory decision layer, never as a replacement for them.

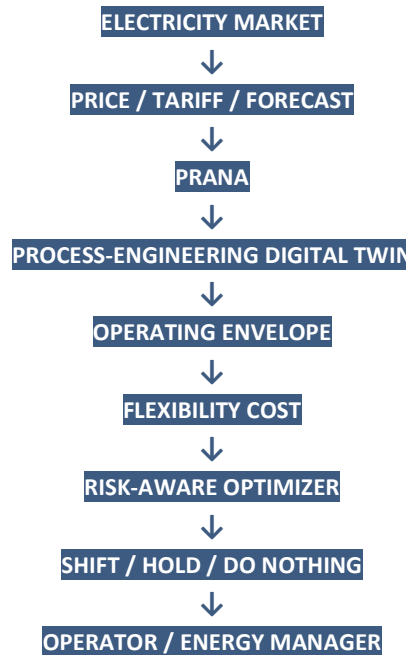


Figure 1. PRANA decision pipeline, market signal to operator recommendation.

3.1 What each block does

- **Market / tariff / forecast** ingests the settled IEX 15-minute panel, the MERC tariff order, and a quantile price forecast for the next delivery day.
- **Process-engineering digital twin** a reduced-order model of the specific asset (chlor-alkali cell house, air separation unit, electrolyser, etc.) that relates production rate to electrical power, inventory, and co-product flows.
- **Operating envelope** the set of hard bounds safety interlocks, ramp limits, inventory limits, contract demand inside which any recommendation must fall.
- **Flexibility cost** the $\phi(\Delta MW, \Delta t)$ curve (\$5): what it costs the process itself to move load.
- **Risk-aware optimizer** a stochastic mixed-integer program that weighs the landed-cost saving against the flexibility cost and a risk term, across a range of forecast prices, not a single point forecast.
- **SHIFT / HOLD / DO NOTHING** the only three outputs. DO NOTHING is a frequent, valid, and intentionally unglamorous result: PRANA does not manufacture flexibility merely because electricity is expensive.

4. The Process-Engineering Digital Twin

The word "twin" is used here in a specific, narrow sense and not as a synonym for a 3-D visualisation or a general-purpose plant simulator. Industrial process behaviour that is relevant to electrical flexibility spans several disciplines simultaneously:

- chemical / process engineering reaction stoichiometry, co-product ratios, mass and energy balance;
- electrical engineering rectifier and drive behaviour, power factor;
- mechanical / equipment engineering ramp limits, cycling wear, single-train derates;
- utilities steam, compressed gas, cooling balances that move with production rate;
- control engineering the setpoints and interlocks that actually govern what a plant is allowed to do;
- inventory and safety buffer capacity, minimum cover, licensed operating limits.

PRANA's twin is therefore a **reduced-order, decision-focused** model built from published process-engineering relationships for the specific archetype, not a generic curve fit. It is designed to answer one question only:

"How much electrical load can this process safely and economically move, for how long?"

It is explicitly **not** a replacement for a distributed control system (DCS), a safety instrumented system (SIS), advanced process control / model-predictive control (APC/MPC), or full-fidelity process simulation software such as Aspen or HYSYS. Those systems continue to govern the plant. PRANA's feasible set is designed to always be a strict subset of what those systems already permit every recommendation must additionally clear the plant's own interlocks, which is why PRANA is positioned as advisory (§19) rather than as a control-layer replacement.

5. The Flexibility Cost Model $\phi(\Delta MW, \Delta t)$

PRANA's central engineering object is a *flexibility cost curve*, written $\phi(\Delta MW, \Delta t)$:

$$\phi(\Delta MW, \Delta t) = \text{estimated process/economic cost, in ₹ per MWh shifted, of moving electrical load by } \Delta MW \text{ for a duration } \Delta t$$

where ΔMW is the electrical load movement requested and Δt is the duration for which it is held. ϕ allows PRANA to compare the electricity-market benefit of moving load against the process's own cost of moving it, before recommending anything.

5.1 Components actually implemented for the chlor-alkali twin

Only components present in the supplied chlor-alkali implementation are listed here; no additional implemented cost terms are claimed.

- Round-trip electrochemical loss because electrical power is convex in production rate (\$6), splitting production into a turn-down leg and a rebuild leg always consumes more kWh than holding steady output, even though total product is unchanged. This convexity-driven loss is the physical core of ϕ .

- Membrane cycling cost charged per tonne/hour of setpoint travel, standing in for accelerated membrane wear from movement rather than runtime (assumption, §21, item A-07).
- Diverted co-product cost chlorine that cannot be consumed downstream and is routed to the bleach plant is valued at a stated rupee-per-tonne rate rather than treated as free.
- Power-factor penalty rectifier power factor droops on turn-down (thyristor firing-angle behaviour); the resulting extra kVAh is charged at the block's own landed rate.
- Terminal-inventory / shortfall penalty ending the delivery day short of committed inventory is charged at a level high enough to prevent it from ever being a deliberate arbitrage.

No production penalty, alternative-supply cost, or recovery cost beyond these is implemented for the chlor-alkali twin. ϕ lets the optimizer compare the electricity benefit of a proposed move against the sum of these process costs, and decline the move (DO NOTHING) whenever the process cost is not covered by the electricity saving.

6. Flagship Demonstrator: The Chlor-Alkali Cell House

[MODELLED REFERENCE CASE] The 700 TPD NaOH plant analysed in this section is a modelled reference archetype. No specific, named plant's historian has been fitted to it.

6.1 The archetype

700 tonnes/day NaOH (100% basis), 67 MW nominal connected electrical load, membrane-cell technology, with lye (caustic) storage tankage sized in the range 300–1,800 t. Scale mostly cancels in the ₹/kWh headline figure quoted in §14; it does *not* cancel in the absolute rupee-crore figures, which are specific to this archetype's size.

6.2 Electrochemical basis derived, not fitted

The National Productivity Council (Government of India) publishes the governing relationship for a chlor-alkali cell (Chlor-Alkali Sector Manual, p.28): operating cell voltage rises almost linearly with current density,

$$V = V_0 + k \cdot i$$

Because production rate is proportional to current (Faraday's law), and power is voltage × current, this relationship forces electrical power to be a *convex, quadratic function* of production rate the twin does not assume this shape, it follows from the cited electrochemistry:

$$P(x) = P_{nom} \cdot (a + b \cdot x + c \cdot x^2)$$

with $a = 0.06$, $b = 0.73$, $c = 0.21$ (x is production rate as a fraction of design). The National Productivity Council's own empirical cell curve (p.29), $E = 2.41 + 0.329 \cdot i + 0.24 \cdot \log i$, implies a quadratic coefficient of 0.30–0.45 across the operating range steeper than the 0.21 used here. **The twin therefore understates the round-trip loss of flexibility, in the conservative direction: re-running the headline result with the NPC-implied coefficients moves it up by roughly 6% on a 45-day sample (₹10.90 → ₹11.55 crore/year).** The self-consistency test on this curve fits against a self-chosen voltage law and pins internal consistency only, not correctness against a real cell this is stated openly rather than presented as validation.

A further, less obvious consequence of the same equation, worth stating because it is counter-intuitive: specific energy consumption (kWh per tonne) falls as the cell turns down. Chlor-alkali is, in this narrow energetic sense, a process that is paid to be flexible rather than penalised for it.

6.3 The co-product constraint

Chlorine is produced simultaneously with caustic soda, in a fixed stoichiometric ratio (NPC cites 1:0.89). Chlorine is a Schedule-3 hazardous substance under Indian rules and is not stored in bulk in practice it must be consumed essentially as it is made. This means the cell house cannot turn down further than its downstream chlorine consumer allows, *regardless of what the cell's own safety interlock would otherwise permit.*

6.4 Which constraint actually binds

Two turndown floors exist in the model, and only one is active in practice:

Constraint	Floor	Status	Source
Cell H ₂ -in-Cl ₂ safety interlock	40% of design	Modelled as a hard bound	Assumption no source found at any tier (§21, A-03)
Chlorine consumer's minimum offtake	70% of design (base case)	BINDING this is the floor that is actually reached	Assumption the single most load-bearing assumption in the model (§21, A-04)

Table 1. The economically deliverable flexibility is set by the co-product / downstream system, not by the cell's own safety interlock. This finding must not be generalised to every chlor-alkali plant; it depends on this archetype's assumed downstream configuration.

This is a plant-configuration-specific finding for this archetype, and is not presented as a general property of chlor-alkali plants. A plant with a merchant chlorine offtake, a different bleach-plant capacity, or a different downstream customer mix would have a different binding constraint.

7. Chlor-Alkali Process Constraints Modelled

The optimizer is bound by the following constraint set for the chlor-alkali archetype. Status labels indicate whether a value is grounded in a cited external source, is a stated engineering assumption, or is inactive in the modelled schedules.

Constraint	Modelled value	Status
Production range	40–108% of design (cell), 70–105% (consumer draw)	Cell range: assumption; consumer range: assumption (load-bearing)
Power–production curve	$P = P_{nom}(0.06 + 0.73x + 0.21x^2)$	Coefficients: conservative vs. NPC-implied (grounded, §6.2)
Chlorine balance	1:0.89 co-product ratio	Grounded NPC p.20
Chlorine consumer minimum take	70% of design draw (base case)	Assumption no source found; swept 60–90% (§21, A-04)
Bleach-plant diversion (sink of last resort)	Capacity-limited, valued at ₹9,000/t	Assumption
Caustic inventory buffer	20.5 h nominal cover, 1,500 t usable	Assumption likely overstated 3–5×, but not binding: measured schedules use only 4.2 h (8% of the modelled band)
Ramp rate	60%/hour (0.15 of design per 15-min block)	Assumption no source at any tier; most likely-to-be-wrong parameter; not swept (§21, A-05)
Membrane cycling cost	₹1,200 per t/h of setpoint travel; 60 t/h/day movement budget	Assumption inside an assumption (§21, A-07)
Power factor	0.98 → 0.85 linear to minimum load, 4 rectifier trains	Real behaviour follows $\cos \alpha$ and is not linear; disclosed
Contract / peak billing demand	MD ratchet at 75% of contract demand	Not found in the operative MERC order (§21, A-12)
Terminal inventory shortfall	₹50,000/t penalty	Assumption, chosen to prevent arbitrage; verified 0.00 t shortfall on normal days

Table 2. Chlor-alkali process constraints as modelled. Full basis and direction of error for every assumption is in §21 and the project's ASSUMPTIONS.md.

8. Electricity and Landed Cost Model

PRANA does not optimize against the exchange clearing price (MCP) alone, because the exchange price is not what an industrial consumer pays at the meter. The **landed cost** is built as:

$$\text{Landed cost} = \text{MCP} + \text{wheeling} + \text{cross-subsidy surcharge} + \text{additional surcharge} + \text{SLDC charge} + \text{duty} + \text{demand charge} + \text{power-factor effect} + \text{ToD multiplier}$$

8.1 The verified tariff core

[REAL MARKET DATA] The following figures are read directly from a primary regulatory document: MERC Order, Case No. 75 of 2025, post-remand proceedings, dated 25 March 2026 (page 100, “Summary of HT Tariff for FY 2026-27”, category HT I(A) HT-Industry, EHV connection). The earlier 25 June 2025 order was quashed and set aside by the Bombay High Court and must not be cited; the Supreme Court remanded the matter and the post-remand order is the operative instrument used throughout.

Parameter	FY2026-27 value	Note
Demand charge	₹650/kVA/month	EHV connection; p.100
Energy charge	₹8.44/kVAh	Billed per kVAh, not kWh a ~2% unit trap at 0.98 PF
Demand charge, daily marginal	₹22,109 per MW of peak, per day	Derived: ₹650 ÷ 0.98 PF × 1000 kVA/MW ÷ 30 days
ToD peak surcharge	+20%, 17:00–24:00 (7 hours)	Table 8, p.77; a footnote marker on this row is unresolved in the 123-page order disclosed as a known ambiguity
Solar-hour rebate	–15% Apr–Sep, –25% Oct–Mar	FY2026-27; escalates to –20%/–30% from FY2028-29
Night rebate	REMOVED flat 0%, 00:00–09:00	§18.13: regulator states this is because a night rebate competed with shifting load to solar hours
Wheeling (33 kV)	₹0.81/kVAh	Nil at EHV

Table 3. Verified MERC tariff parameters. An earlier draft of this project used a placeholder demand charge of ₹590/kVA/month before the primary order was read; that figure is superseded and must not be quoted.

A forward-looking point worth making explicitly: on the regulator's own published multi-year trajectory, the demand charge rises from ₹650 to ₹750/kVA/month (FY2026-27 → FY2029-30, +15%) while the energy charge falls from ₹8.44 to ₹7.45/kVAh (–12%). Flexibility that manages peak demand becomes **more** valuable on this trajectory, not less and this is read from the primary order rather than forecast.

8.2 What remains a placeholder

[ENGINEERING ASSUMPTION] The open-access surcharge stack wheeling, cross-subsidy surcharge, additional surcharge, and SLDC charge used in the EXCHANGE-mode headline result are placeholders (wheeling ₹1.08, cross-subsidy ₹1.31, additional surcharge ₹0.00 assuming a waiver, SLDC ₹0.06/kWh, duty 9.3%). These are collectively about 46% of the landed cost claim and are the largest single unquantified risk in the economics presented in this document. If the additional-surcharge waiver assumed here does not apply in practice (raising it to roughly ₹1.39/kWh), the open-access economic case can invert. This is disclosed here rather than left for a reviewer to find.

9. Market Data

[REAL MARKET DATA] IEX day-ahead (DAM), real-time (RTM), and green day-ahead (G-DAM) markets, Maharashtra bid area (W2), 15-minute settlement blocks, 2022-04-01 to 2026-08-06: 152,542 blocks, 1,588 complete days.

This is real, settled market data — actual money changed hands at these prices. No plant-side data is mixed into this panel; the market panel and the modelled plant are cleanly separated throughout the platform.

9.1 Preprocessing and price-cap handling

Regulatory price caps changed over the panel period (₹20,000 → ₹12,000 → ₹10,000/MWh, April 2022 / May 2022 / April 2023) and are handled as empirically derived regimes rather than a single constant; 5,372 RTM blocks cleared above ₹10,000/MWh in the pre-April-2023 period and are treated under the regime applicable at the time. The deviation settlement (DSM) rate used for risk-constraint purposes is modelled as approximately $\max(\text{DAM}, \text{RTM})$ for the same block, based on an audit of 133,056 WRPC deviation blocks. *That audit lives in a separate project (a related thesis) and is not reproducible from this repository directly* it is treated in this document as cited-but-not-shipped rather than as an independently verifiable claim within this submission.

9.2 Real data versus modelled plant data — kept strictly separate

Data category	Status	Example
Market prices (IEX)	REAL	152,542 settled 15-minute blocks
Tariff structure (MERC core)	REAL	Demand charge, energy charge, ToD table
Tariff structure (open-access stack)	ASSUMPTION	Wheeling, CSS, additional surcharge, SLDC
Cell voltage law	REAL (published)	NPC Government of India manual, p.28–29
700 TPD plant, its historian, its actual operating data	MODELLED does not exist	Every rupee-crore figure in §14 derives from this
Turndown floor, ramp rate, buffer size, cycling cost	ASSUMPTION	§21 — each with a stated basis or an explicit absence of one

Table 4. Data provenance, kept explicit throughout this document.

10. The AI / Machine-Learning Component

PRANA forecasts next-day prices with a **LightGBM quantile regression** model producing three quantiles per 15-minute block — q10 (low case), q50 (middle case), q90 (high case) — from 15 causal features, the strongest of which is the same-block day-ahead price, known 12–36 hours ahead. Gradient-boosted trees were chosen over a deep sequence model (e.g. an LSTM) deliberately: they are the established strong baseline on tabular electricity-price data, train in minutes, and — as §11 shows — forecast skill is a comparatively small lever in the total value PRANA creates, so the build effort was not spent optimizing the least important component.

10.1 Documented performance

Metric	Value
Forecast MAE	1,072 ₹/MWh
vs. naive persistence	1,189 ₹/MWh
vs. day-ahead price as forecast	1,271 ₹/MWh
vs. 7-day block-mean	1,538 ₹/MWh
80% interval coverage, raw	69.7%
80% interval coverage, after out-of-sample conformal widening	76.2% (target 80%)
Train / test split	Trained through 2026-03-31; tested 2026-04-09 → 2026-08-06 (no leakage)

Table 5. Forecast performance, all figures from the model's own evaluation code.

The forecaster is under-covered relative to its 80% target (76.2% achieved) the test period is more volatile than the calibration window. This is reported rather than tuned away, and it means the risk term in the optimizer (§11) is mildly optimistic; no confidence-interval correction beyond conformal widening has been applied.

10.2 What AI does, and does not, decide

The forecaster predicts a *range* of prices, with uncertainty. It does not decide feasibility engineering constraints (§4, §7) do that and it does not make the final economic call; the optimizer (§11) does. This separation is deliberate: a forecast is a statistical estimate of an unknown future price, and is never allowed to override a hard safety or process bound.

11. Stochastic Optimization

The dispatcher is a 96-block (15-minute resolution, one full delivery day), stochastic mixed-integer linear program (MILP), solved with the open-source CBC solver. A conceptual formulation, without reproducing the implementation:

$$\min E[\text{landed electricity cost}] + \lambda \cdot \text{CVaR}_{95}[\text{cost}] + \text{flexibility cost } \varphi + \text{cycling cost} + \text{demand charge} + \text{shortfall / diverted-product penalties}$$

subject to:

- process operating envelope (cell range, ramp rate, power-factor behaviour),
- caustic inventory balance and buffer limits,
- chlorine co-product balance and the consumer's minimum take,
- contract / billing demand,
- and any constraint supplied by the constraint agent (§12).

The curved power relationship $P(x)$ (§6.2) cannot be expressed directly inside a mixed-integer *linear* program. It is approximated by **16 tangent hyperplanes** straight-line segments fitted to sit just under the true curve and the resulting linearisation error is measured, not assumed: **0.025% for the chlor-alkali twin** (0.005% for the air separation unit twin, 0.023% for the electrolyser twin, 0.622% for the pipeline twin). Risk is priced through a CVaR₉₅ term (Rockafellar–Uryasev): rather than optimizing only the expected cost

across the forecast's q10/q50/q90 fan, the objective also penalises the worst 5% of outcomes, so the schedule does not fall apart when the forecast is wrong.

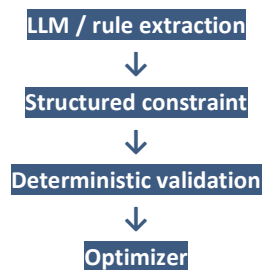
Deviation from the schedule (settling the gap between plan and actual at the market's deviation-settlement rate) is offered to the optimizer as a decision variable in every solve and is **provably never selected**: because the published deviation rate is modelled as approximately $\max(\text{day-ahead, real-time})$ price, it is never arbitrage-favourable. **Deviation used: 0 MW in every reported solve.**

12. The Constraint Agent

A large-language-model agent converts operator language and standard operating procedures into structured, machine-checkable constraints. For example, the sentence

"Keep tank inventory above 500 tonnes."

becomes the typed constraint $\text{inventory} \geq 500 \text{ t}$, which is then validated deterministically before being handed to the optimizer:



The LLM is never permitted to control the process directly, and this is the architectural decision that makes PRANA's explanations trustworthy: the language model does not generate the *reasons* a schedule was chosen. Every explanation is derived from the solver's own binding constraints (which bounds are active in the solution); the LLM only phrases those solver-derived facts in plain language. It is never asked to invent a justification, so it cannot. Human sign-off remains required for every constraint before it affects a live schedule.

13. Decision Output

PRANA returns exactly one of three recommendations per block, never a continuous dial disconnected from feasibility:

Output	Meaning
SHIFT	The process can absorb a load movement for less than the market is paying for it. Recommended.
HOLD	Flexibility exists in principle but its process cost (ϕ) exceeds the market saving. Not recommended at this price.
DO NOTHING	The flexibility does not exist at the requested depth or duration ϕ is effectively infinite. Never touch the process here.

DO NOTHING is a valid, frequent, and intentionally important output. PRANA does not force flexibility merely because electricity is expensive at a given moment; it first asks whether flexibility is *available*, then

whether it is *safe*, then whether it is *economically worthwhile*. Only if all three hold does it recommend a move.

14. The 120-Day Backtest

[MODELLED REFERENCE CASE] 120 consecutive modelled delivery days, replayed against real, settled IEX prices, on the modelled 700 TPD chlor-alkali archetype. This is a simulation on real prices, not a measurement from an operating plant.

14.1 Headline result

Metric	With forecast error (headline)	Perfect foresight
Saving per kWh of total plant load	₹0.133	₹0.174
Saving per day	₹2.51 lakh	₹3.31 lakh
Annualised	₹9.17 crore	₹12.07 crore
As % of the electricity bill	1.53%	2.01%
Days worse than doing nothing	13 of 120	0
Constraint violations (independent check)	0	0

Table 6. Headline 120-day backtest. Quote the forecast-error column, not the perfect-foresight column, as the representative figure.

14.2 Gross to net the calculation shown explicitly

The gross benefit is netted of every cost the flexibility itself creates, over the same 120 days:

Item	₹ lakh, over 120 days
Electricity avoided (gross benefit)	378.3
less: membrane life consumed by setpoint movement, chlorine diverted to the bleach plant, and extra kVAh from power-factor droop	-76.8
Net benefit	301.5

Table 7. Gross-to-net calculation. About one fifth (20%) of the gross saving is paid straight back to the plant as wear and diverted product; this is disclosed rather than netted out silently.

Annualisation, shown explicitly

$$₹301.5 \text{ lakh} \times (365 \div 120) \approx ₹917.06 \text{ lakh} \approx ₹9.17 \text{ crore / year}$$

14.3 Additional documented results

- The forecast-error haircut comparing the headline (₹0.133/kWh) against perfect foresight (₹0.174/kWh) is approximately 24% ($1 - 0.133/0.174$). The project's own slide deck and playbook round this to “22%” in places; both describe the same comparison and the discrepancy is a rounding difference, flagged here for consistency.
- Deviation used: 0.000000 MW in every solve.
- Constraint violations against the true nonlinear physics: 0, across all 120 days (§15).
- The demonstration day used in the live-solve demo (2026-07-12, perfect foresight) saves 1.72× the 120-day mean and is not representative of a typical day; this is disclosed wherever that day's figures are shown (§Appendix C).

[MODELLED REFERENCE CASE] “₹9.17 crore/year” is an annualised modelled economic potential for the reference archetype. It is NOT measured savings from an operating plant.

15. Independent Physics Verification

The optimizer's internal representation of the power curve uses the 16-segment tangent-hyperplane linearisation (§11) because a MILP requires linear constraints. This linear approximation is not trusted as the final check. A separate verification step, run after every solve, *recomputes power from the exact convex (quadratic) curve*, re-integrates the caustic inventory block by block using that exact power, and re-checks every modelled bound the power curve, ramp rate, inventory, co-product balance, and contract demand.

A schedule that satisfies only the linear approximation but would violate the true nonlinear physics is reported as a violation by this step. Across all 120 backtest days, the result was **zero violations**. This matters because it is the difference between “the optimizer's internal model is self-consistent” and “the schedule the optimizer proposes is actually feasible against the real physical relationship” the second is what an engineer signing off on a schedule needs to know.

16. Baseline Comparison What a Simple Rule Already Captures

The obvious objection to any optimizer is: *couldn't a simple, production-neutral rule do most of this?* PRANA is compared against exactly that rule: shed load through the 17:00–24:00 tariff-peak window, run at the deliverable maximum during the solar window, and solve the depth of shedding so that the day's total production balances the way an experienced operator would write it on a whiteboard. Both PRANA and the rule are settled on identical terms, including the terminal-inventory charge.

	PRANA (full optimizer)	Time-of-day rule
Saving, 60-day sample	₹283,219/day	₹50,881/day
Share of PRANA's saving captured	100%	18%
Days the rule loses money		15 of 60

Table 8. The simple rule captures 18% of PRANA's value; PRANA's incremental value over the rule is approximately ₹8.48 crore/year.

The structural reason: a fixed rule must keep production neutral *within* the day, which caps how far it can turn down (roughly 91% of design in this sample). The optimizer can move inventory *across* days and respond to the actual price movement *inside* a tariff slab, capturing spread that a coarse four-step ToD tariff prices at zero.

A correction carried forward openly

An earlier version of this comparison omitted the terminal-inventory charge for the simple rule, and the rule appeared to beat PRANA by 382% an artefact of a rigged comparison, not a real result. It was found, fixed, and is recorded here (and in the project's own bug log, §24) so the error is not repeated or hidden.

17. Multi-Industry Generalisation

The architecture is designed so that different process-engineering physics plug into a common flexibility-intelligence pipeline (§3). Five archetypes exist in the codebase today, at clearly different levels of engineering maturity, and none is presented at a higher tier than it has earned:

Archetype	Maturity	Basis
Chlor-alkali cell house	Flagship quantified, 120-day demonstrator	NPC Government of India manual (grounded convexity, SEC band, co-product ratio, BoP share)
Air separation unit (ASU)	Grounded at the same tier as the flagship; not independently backtested in this document	Peer-reviewed literature: Pattison, Touretzky, Johansson, Harjunkoski & Baldea; Zhang, Grossmann, Heuberger, Sundaramoorthy & Pinto; Caspari et al., AIChE J. 2019; Cao et al., AIChE J. 2017. PRANA's 55% floor matches the standard-design flooding bound
Electrolyser	Technology-class grounded; one parameter flagged optimistic	SEC of 50 kWh/kg is within the <50 kWh/kg nominal technology-class range; the modelled 15% minimum-load floor is below the ~30% comfortable range cited in the literature and is likely optimistic
Refinery utility block	Engineering demonstrator coded, ungrounded	No public source found for its parameters. No rupee figure from this twin is presented as validated in this document (§18)
Crude/product pipeline	Engineering demonstrator coded, ungrounded	No public source found for its parameters; excluded from earlier targeting after an expert review found the model did not represent the actual limiting variable (temperature) for that asset

Table 9. Archetype maturity, stated plainly. Only chlor-alkali carries an economic figure that is presented as this document's flagship number.

The common element across archetypes is not the physics which changes completely from a cell house to a cryogenic column to an electrolyser but the pipeline: derive a convex power–production relationship from the asset's own governing physics, quantify the operating envelope, price the flexibility, and dispatch it against landed cost under uncertainty.

18. Application to Refinery Operations

A refinery panel is expected to bring specific scrutiny to this section, so it is stated conservatively and explicitly.

PRANA does not claim, and this document does not present, a complete refinery digital twin. A refinery utility-block twin exists in the codebase and has been run in earlier exploratory work, but **its parameters are not grounded against any public or plant-specific source and no rupee saving figure from it is presented as a validated claim in this document.** An expert review during this project's development separately found two structural reasons a refinery is a weaker fit for this approach than chlor-alkali, and the project changed its flagship target as a result rather than defending the original pitch: refineries

typically run captive power plants, so the marginal cost of grid electricity is close to flat and the arbitrage this platform monetises is much smaller; and a refinery's air separation unit is commonly built, owned and operated by a third-party industrial-gas company under a take-or-pay contract, so the refinery may not even control the asset that would need to flex.

18.1 What a real refinery deployment would require

A genuine deployment would build process-specific reduced-order models for the subsystems actually capable of safe electrical flexibility potentially pumps, compressors, utility systems, hydrogen balance, steam balance, and cooling systems with the same objective used throughout this document:

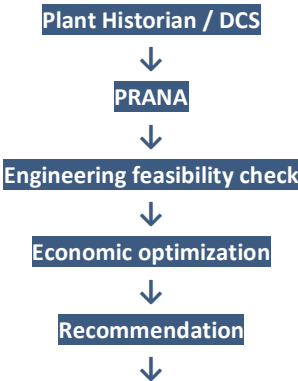
determine safe electrical flexibility while preserving throughput, pressure, hydrogen balance, steam balance, equipment operating envelope, anti-surge margins, downstream requirements, and safety.

PRANA does not tell a refinery to sacrifice production for an electricity saving. It is designed to identify flexibility only within the plant's existing, already-licensed operating envelope the same design principle applied to the chlor-alkali flagship (§6.4).

19. Real-World Deployment Architecture

Stage	Description
1	Offline engineering model twin fitted to published/engineering-range parameters (current state of this submission)
2	Historian integration twin re-fitted to a specific plant's actual energy, production and inventory data
3	Read-only shadow mode PRANA produces a recommended schedule daily; no control action is taken; performance is compared against actuals
4	Engineer validation the plant's own engineers review shadow-mode recommendations against SOPs and interlocks
5	Pilot a bounded, supervised trial with human sign-off on every recommendation before it is acted on
6	Controlled operational integration only if the pilot succeeds and the plant approves; scope and authority defined by the plant, not by PRANA

Table 10. Six-stage deployment path. No stage after 3 is claimed to exist for any real plant today.



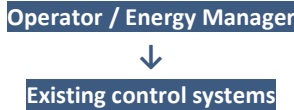


Figure 2. PRANA is an advisory decision layer. It sits alongside, and does not replace, the DCS, APC/MPC, or safety instrumented systems already governing the plant.

20. Data Requirements for Real Deployment

A real site considering calibration would need to provide the following. This list distinguishes what a real deployment would require from what the current prototype already has.

Data needed for calibration (not yet available)	Already available in this prototype
15-minute or higher-resolution plant electricity meter data	Real 15-minute market price panel (152,542 blocks)
Process production records	Published cell-voltage law and empirical cell curve (NPC)
Equipment load / rectifier historian data	Verified MERC tariff core (demand charge, energy charge, ToD table)
Process variables (current, voltage, tank level)	A working, tested optimizer, forecaster, and verifier (§14–15)
Downstream / co-product consumer demand data	A regression-tested software platform (§23)
Utility balances relevant to the specific plant	
Equipment operating envelopes and interlock schedule	
Full tariff structure applicable to the specific connection	
Historical operating states / SOPs / alarm and constraint records	
Maintenance information relevant to cycling cost	

Table 11. Calibration data gap. Roughly 30 days of read-only historian access to one plant is the stated minimum to begin closing this gap (§21, item A-14).

21. Assumptions and Limitations

Every number in this document that is not measured or published is an assumption, and every assumption is listed here with its basis and its stated direction of error, drawn directly from the project's own ASSUMPTIONS.md audit. *The test of a model is not whether it has assumptions; it is whether they can be listed, their direction of bias stated, and their effect on the answer shown when they move.*

Parameter / component	Status	Implication
700 TPD plant archetype	Does not exist as a specific plant	₹/kWh headline mostly scale-invariant; absolute ₹ crore figures are archetype-specific
Power-curve coefficients (a=0.06, b=0.73, c=0.21)	Conservative vs. NPC-implied 0.30–0.45	Re-running with NPC coefficients moves the headline up ~6%
Cell turndown floor (40%)	No source at any tier; superseded in practice by the chlorine-consumer floor	Not the active constraint in any reported schedule
Chlorine consumer's turndown (70% base case)	No source; single most load-bearing assumption	Swept 60–90%; value nearly flat 60–80%, breaks only at 90%
Ramp rate (60%/hour)	No source at any tier; the parameter most likely to be wrong, and almost certainly too fast	If the true rate is 10–15%/h, the schedule mis-times the shed and delivers materially less on day one; not swept
Caustic buffer (20.5 h cover)	No source; likely overstated 3–5×	Robust by accident: measured schedules use only 4.2 h (8% of the band)
Membrane cycling cost (₹1,200/t·h)	An assumption inside an assumption (a ~₹21.5 cr membrane set, 5% life consumed by cycling over 4 years)	Contributes to the 20% of gross saving given back
Power factor curve (0.98→0.85 linear)	Real behaviour follows $\cos \alpha$, not linear	In EXCHANGE mode (the headline configuration) this penalty returns zero disclosed, not hidden
Open-access surcharge stack (wheeling, CSS, additional surcharge, SLDC)	All placeholders; ~46% of the landed-cost “moat” claim	If the assumed additional-surcharge waiver does not apply, the open-access case can invert
DSM $\approx \max(\text{DAM}, \text{RTM})$	Audit is real (133,056 WRPC blocks) but lives outside this repository	Cited but not independently reproducible from this submission
MD ratchet (75% of contract demand)	Not found in the operative order (a review order, not a full tariff schedule)	A lower ratchet would increase PRANA's value
Grid emission factors (0.90 / 0.55 tCO ₂ /MWh)	Approximate; no marginal factor sourced	No CO ₂ figure is quoted anywhere in this document
Sector extrapolation (≈₹155 crore/year)	One archetype's ₹/kWh × installed capacity; assumes universal open access, storage and flexible offtake	Order-of-magnitude only; never defended as an estimate
ASU / electrolyser / refinery / pipeline twin parameters	ASU and electrolyser grounded at technology-class level; refinery and pipeline ungrounded, no public source found	No rupee figure presented for refinery or pipeline twins in this document
No live plant historian, no real-time control integration	Stage 1 of the six-stage deployment path only (§19)	Calibration requirement before any real deployment

Table 12. Full assumption register. Basis and stated direction of error for each item mirrors the project's own ASSUMPTIONS.md.

21.1 Known limitations, stated plainly

- The tariff/market layer is validated on real settled data but has never been run live against an actual utility bill.
- The process-twin layer is an analytical proof of the critical characteristic via a published NPC relationship, with no experimental proof against any real plant.
- The backtest covers 120 days, one bid area, one price regime, and one archetype in depth. The best 20 days carry 31% of the total modelled saving.
- The demonstration day (2026-07-12) saves 1.72× the 120-day mean and is not typical.
- No confidence intervals are placed on any headline figure.

22. Novelty and Innovation Positioning

PRANA does not claim to have invented industrial demand response, process flexibility, ramp constraints, inventory-based storage framing, virtual batteries, or model-predictive control. Chlor-alkali demand response specifically is an established, active research area treating it as an open problem would misrepresent the field to anyone who follows it.

22.1 The published field, and where its numbers sit

Otashu & Baldea (University of Texas at Austin) built demand-response-oriented dynamic models and scheduling frameworks for membrane chlor-alkali plants. Weigert, Hoffmann, Esche, Fischer & Repke (TU Berlin) produced the first such dynamic model validated against real industrial plant data. Richstein & Hosseinioun (DIW Berlin) modelled the chlor-alkali process with a storable intermediate good against network tariffs. Published electricity-cost savings in this literature cluster around 4% (I&EC Research, 2020) to 5.8% (Applied Energy, 2023) of the electricity bill under time-dependent tariffs; profit improvement reaches roughly 10% when reserve-market participation is included. **PRANA's 1.53% of the electricity bill sits below this entire published range**, for two structural, disclosed reasons: India's demand charge on peak billing kVA penalises exactly the peak that load-shifting creates, and the chlorine consumer's turndown binds the shed to 70% of design in this archetype. This comparison's denominator also differs PRANA's is the full landed bill, while several reviewed papers use energy or wholesale cost alone, which mechanically inflates their percentage relative to PRANA's; this is stated as a caveat on the comparison, not smoothed over.

Weigert et al. (2021), whose model is independently validated against a real plant, state that chlor-alkali electrolysis “cannot be operated flexibly by itself as storing chlorine is avoided in practice due to safety concerns” the same structural conclusion this project reached independently and encodes as the chlorine consumer's minimum-take constraint (\$6.3–6.4).

22.2 What is actually presented as new

- A real, four-year settled Indian price panel (152,542 blocks, IEX Maharashtra W2) the reviewed literature is predominantly European or US and largely uses exchange or stylised prices.
- A primary Indian regulatory tariff order opened to the specific page (MERC Case 75 of 2025, post-remand, p.100), including establishing that a widely circulated earlier order was quashed.

- A landed-cost engine ToD multipliers, open-access surcharges, duty and the demand charge on peak billing kVA dispatched inside the optimization itself rather than applied as a post-hoc adjustment; no reviewed paper prices a full Indian regulatory stack this way.

Not claimed as new: the buffer-as-battery framing, the convex power curve itself, demand-response scheduling of chlor-alkali in general, or the observation that demand charges impede flexibility. All predate this work and are cited above.

23. Engineering Validation and Reproducibility

Evidence	Status
Automated regression tests	45 tests, each pinning a specific claim made in the project's documentation
Demo execution	Runs offline from a clean clone; the terminal demo reproduces every headline figure in under 30 seconds
Market data	152,542 real settled IEX blocks, self-contained after first build
Source traceability	Every number in the project's pitch material is tied to a primary document, a script, or an explicitly labelled assumption
Documented assumptions	Full register with basis and direction of error (§21)
Documented bugs and fixes	§24

Table 13. Evidence of engineering discipline. A note on numerical consistency: the instructions accompanying this submission's source material state "48/48 tests passing", while the project's own README, source-of-truth deck, and audit trail consistently report 45 regression tests. This document reports 45, the figure independently corroborated across the project's own artefacts, and flags the discrepancy here rather than silently choosing either number.

None of this evidence is presented as commercial validation. It demonstrates that the software runs correctly against its own internal claims and against the true nonlinear physics (§15); it does not demonstrate that the modelled plant behaves like any specific real plant, which is precisely the gap Stage 2 of the deployment path (§19) exists to close.

24. Development History Errors Found and Corrected

Presented here because finding and correcting these errors, and recording them rather than quietly fixing them, is itself part of the engineering case for this platform.

Bug	Impact	Fix	Lesson
Infeasible-solve bug	An outage was reported as a ₹165,121 saving	Solver status is now gated before any figure is printed	Never trust an optimizer's output without checking its solve status first
Horizon-scaling bug	A spurious +317% multi-day value	Recognised the demand charge as a daily, not per-block, rate; corrected to +16.2%	Charges with a different time basis than the optimization horizon must be handled explicitly
Time-of-day baseline bug	The simple rule appeared to beat PRANA by 382%	The terminal-inventory charge, omitted for the rule, was added on identical terms to both	A counterfactual must be settled on exactly the same terms as the proposed system

Bug	Impact	Fix	Lesson
Withdrawn battery-size claim	41 MW / 1,410 MWh, quoted against the cell's 40% interlock rather than the chlorine consumer's 70% floor	Corrected downward to 21.8 MW / 748 MWh the deliverable figure at the constraint that actually binds	Quote the binding constraint, not the most flattering one

Table 14. Selected development history. Each fix corresponds to a regression test that would fail if the error recurred.

25. Economic Impact

[MODELLED REFERENCE CASE] All figures below are modelled results for the 700 TPD reference archetype, not measured plant savings.

Metric	Value
Net benefit, 120-day backtest	₹301.5 lakh
Annualised	₹9.17 crore/year
As % of the electricity bill	1.53%
Sector-wide extrapolation (order of magnitude only)	≈₹155 crore/year

Table 15. Headline economic figures, annualised from a 120-day modelled replay (§14).

This figure depends on electricity prices, the specific tariff structure applicable to a plant's connection, the process flexibility genuinely available at that plant, the process cost of using it, the plant's exact configuration, and its operating constraints. **It must therefore be recalculated for each real plant before it is quoted to that plant's management.** This document does not imply that every chlor-alkali or refinery plant in India would save ₹9.17 crore a year; it presents the calculation method and the reference-case result together, so that both the number and the way to reproduce it for a different site are visible.

26. Implementation Roadmap

Window	Activity
0–30 days	Site data acquisition and model calibration read-only historian export; fit the twin's parameters to the specific plant, replacing archetype assumptions with plant-specific values
30–60 days	Shadow-mode testingc daily recommended schedule produced and logged; no control action taken; compared against the plant's actual operation and realised prices
60–90 days	Engineer validation and a controlled pilot plant engineers review every recommendation against SOPs and interlocks before any action is taken
After 90 days	Scale to additional assets or processes at the same site, only after the pilot has demonstrated agreement with the plant's own engineering judgement

Table 16. Roadmap. No stage promises autonomous or unsupervised operation; §19's trust ladder (L1 advisory → L2 supervised → L3 bounded closed-loop) governs any eventual move beyond shadow mode, and is a plant decision, not a PRANA default.

27. Conclusion

PRANA does not treat an industrial process as a battery. It treats process flexibility as an engineering asset whose value depends on what the plant can safely afford to change.

The platform's contribution is not any one of its parts in isolation — not the convex power curve, not the landed-cost tariff engine, not the quantile forecaster, not the constraint-extraction agent. It is their integration into a single decision layer that a plant engineer can check, line by line, against the plant's own physics and its own tariff. The chlor-alkali flagship demonstrates this on real market data, a verified primary tariff order, and a published electrochemical relationship — and states plainly, throughout, where the model becomes an assumption rather than a measurement.

Different process physics, plugged into a common flexibility-intelligence framework, is the platform's thesis:

Different process physics + common flexibility intelligence = an industrial energy flexibility platform

What would make this project wrong is stated as plainly as what would make it right: if the true achievable ramp rate is materially slower than the 60%/hour assumed, or if real membrane cells tolerate less cycling than the published guidance implies, the value falls. That is precisely what a 30-day, read-only calibration against one real plant's historian is designed to establish — cheaply, and without touching a control system.

28. References

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- [2] Bombay High Court, Writ Petition 19437/2025 and batch (3 November 2025), quashing and setting aside the Maharashtra Electricity Regulatory Commission order of 25 June 2025; Supreme Court remand order dated 17 November 2025, time extended 9 February 2026.
- [3] National Productivity Council, Government of India, Good Practices Manual - GHG Emission Reduction, Chlor-Alkali Sector, 78 pp. Cell voltage law (p.28), empirical cell curve (p.29), balance-of-plant load and specific-energy-consumption benchmarks by cell generation (p.25, Table 2.7), co-product ratio (p.20), power share of production cost (p.22). npcindia.gov.in.
- [4] Indian Energy Exchange (IEX), Day-Ahead Market, Real-Time Market, and Green Day-Ahead Market settlement data, Maharashtra bid area (W2), 15-minute blocks, April 2022 – August 2026. iexindia.com.
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- [6] Central Electricity Authority (CEA), National Electricity Plan (Volume I - Generation), 2022–32, national storage requirement (73.93 GW / 411.4 GWh by 2031-32, of which battery storage 47.24 GW / 236.22 GWh).
- [7] Alkali Manufacturers Association of India (AMAI), Indian caustic soda installed capacity (64.04 lakh MTPA, March 2025) and capacity utilisation (78.4%, FY2024-25).
- [8] Otashu, J. I. & Baldea, M., “Demand response-oriented dynamic modeling and operational optimization of membrane chlor-alkali plants,” *Computers & Chemical Engineering* (2019); and “Scheduling chemical processes for frequency regulation,” *Applied Energy* (2020).
- [9] Weigert, J., Hoffmann, C., Esche, E., Fischer, P. & Repke, J.-U., dynamic model of a membrane chlor-alkali process validated against real industrial plant data, *Computers & Chemical Engineering* (2021).
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- [11] Pattison, R. C., Touretzky, C. R., Johansson, T., Harjunkoski, I. & Baldea, M., scheduling with low-order dynamic models, air separation unit application (peer-reviewed).
- [12] Zhang, Q., Grossmann, I. E., Heuberger, C. F., Sundaramoorthy, A. & Pinto, J. M., air separation unit scheduling with cryogenic energy storage in energy and reserve markets (peer-reviewed).
- [13] Caspari, A. et al., “Flexible design of air separation units for demand response,” *AIChE Journal* (2019). Operating range 3.5–28 MW for a purpose-designed flexible unit with added storage and refrigeration not applicable to a standard-design ASU.
- [14] Cao, Y. et al., preemptive dynamic operation of air separation units (peer-reviewed), *AIChE Journal* (2017).
- [15] Department of Fertilizers, Government of India, New Urea Policy-2015 (notified 25 May 2015) and amendment notification of 28 March 2018 - Target Energy Norms and normative subsidy mechanism, cited for sector context in the assessment of why fertiliser was not selected as the flagship (§2, §17).

29. Appendices

Appendix A Key Equations

$$V = V_0 + k \cdot i \quad (\text{cell voltage law, NPC p.28})$$

$$E = 2.41 + 0.329 \cdot i + 0.24 \cdot \log i \quad (\text{empirical cell curve, NPC p.29})$$

$$P(x) = P_{nom} \cdot (0.06 + 0.73x + 0.21x^2) \quad (\text{power-production relationship used in the twin})$$

$$\text{Demand charge, daily marginal} = ₹650/\text{kVA/month} \div 0.98 \text{ PF} \times 1000 \text{ kVA/MW} \div 30 \text{ days} = ₹22,109 / \text{MW} / \text{day}$$

$$\text{Annualised saving} = 120\text{-day net benefit} \times (365 \div 120)$$

$$\varphi(\Delta MW, \Delta t) = \text{process/economic cost of moving load } \Delta MW \text{ for duration } \Delta t, \text{ in } ₹/\text{MWh shifted}$$

Appendix B Key Parameters and Provenance

Parameter	Value	Provenance
a, b, c (power curve)	0.06, 0.73, 0.21	NPC BoP share (p.25) + curve shape (p.29); conservative vs. NPC-implied 0.30–0.45
Demand charge	₹650/kVA/month	MERC Case 75/2025, post-remand, p.100
Energy charge	₹8.44/kVAh	MERC Case 75/2025, post-remand, p.100
Chlorine consumer floor	70% of design (base case)	Assumption swept 60–90%
Ramp rate	60%/hour	Assumption unsourced, flagged likely too fast
Market panel	152,542 blocks	IEX, real, 2022-04-01 to 2026-08-06

Appendix C Backtest Calculation Detail

120-day gross-to-net calculation (§14.2):

$$₹378.3 \text{ lakh (gross)} - ₹76.8 \text{ lakh (process costs)} = ₹301.5 \text{ lakh (net)}$$

$$₹301.5 \text{ lakh} \div 120 \text{ days} = ₹2.51 \text{ lakh / day}$$

$$₹2.51 \text{ lakh/day} \times 365 = ₹9.17 \text{ crore / year}$$

Demonstration-day figures (2026-07-12, perfect foresight, not representative saves 1.72× the 120-day mean):

Figure	Steady state	PRANA
Energy	1,896.0 MWh	1,900.5 MWh (+4.5)
Peak billing demand	79.00 MW	82.88 MW (+3.88)
Landed cost paid	₹7.438/kWh	₹7.193/kWh
Day's bill	₹1,41,01,716	₹1,36,70,393
Avoided		₹4.31 lakh (3.06% of day's bill)

The peak billing demand rose by 3.88 MW on this day an explicit, priced trade-off: $3.88 \text{ MW} \times ₹22,109/\text{MW}/\text{day} \approx ₹85,783/\text{day}$ of extra demand charge, accepted because the energy saving exceeded it. This is the reason the demand charge must sit inside the optimization (§8), not be ignored by it.

Appendix D System Architecture

Data (IEX panel, MERC tariff, plant historian when available) → Twin (physics, envelope, ϕ) → Forecast (LightGBM quantiles, conformal calibration) → Decision (stochastic MILP, CVaR_{95}) → Agent (constraint elicitation, solver-derived explanation) → Operator sign-off → Existing plant control systems. Full detail in §3 and §19.

Appendix E Test / Verification Summary

Check	Result
Regression tests	45 passing (see numerical-consistency note, §23)
Independent physics verification, 120 days	0 violations
Tangent-linearisation error, chlor-alkali twin	0.025% (16 tangent hyperplanes)
Deviation used	0 MW, every solve
Forecast interval coverage	76.2% achieved vs. 80% target